DecodeClouds User Guide

How to add a sky photo, identify the clouds, and read the results on decodeclouds.com

At a glance

DecodeClouds identifies the clouds in a photo of the sky. The site is free and needs no account. Using DecodeClouds takes four steps:

1. Open decodeclouds.com.
2. Add a photo by taking one, choosing one, or trying an example.
3. Tap **Identify clouds**.
4. Read, save, or share the results.

The screenshots in this guide show an iPhone. On a computer, the steps are the same, and the words "tap" and "click" mean the same thing. Photos you identify, except the site's examples, join the open DecodeClouds archive; see Privacy and the archive at the end of this guide.

Step 1. Open DecodeClouds

Open Safari on an iPhone, or any web browser on a computer, and go to **decodeclouds.com**.

The first screen shows the Cloud Decoder title and the **Upload a Sky Photo** card (Figure 1).

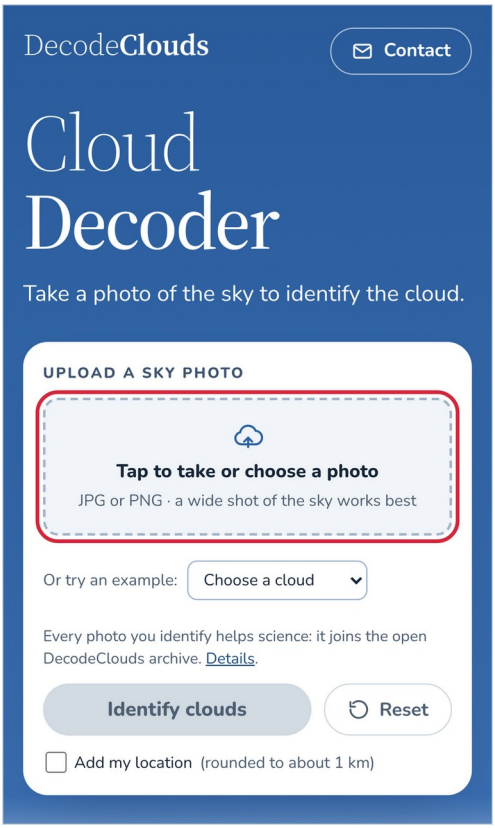


Figure 1. First screen on an iPhone

Step 2. Add a photo

Add a photo in one of three ways. All three ways lead to the same next screen.

Option A, take or choose a photo (phone): Tap the box that reads **Tap to take or choose a photo** (outlined in Figure 1). The iPhone opens its own menu, such as Photo Library, Take Photo, or Choose File. Take a wide shot of the sky, or choose a sky photo already on the phone.

Option B, drag or browse (computer): Drag a photo onto the box that reads **Drag a photo here, or browse**, or click **browse** and choose the photo file (Figure 2).

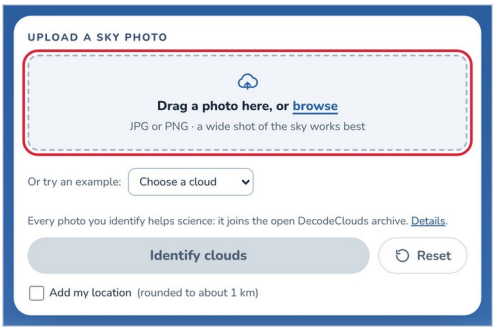


Figure 2. Upload box on a computer

Option C, try an example: Open the **Or try an example** menu, which reads **Choose a cloud**, and pick one of 13 examples (Figure 3). The menu lists the 11 cloud types, Fallstreak hole, and Clear sky.

The examples are the site's own photos, and they are not added to the archive.

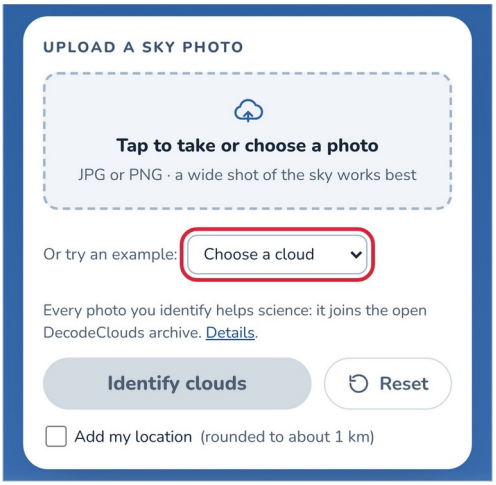


Figure 3. The example menu

The site takes JPG or PNG photos. A wide shot of the sky works best.

Step 3. Check the photo and the options

The photo appears in the Upload a Sky Photo card. For a photo from an iPhone, the line under the photo reads **Photo ready** and the photo's size. For a photo from a computer, the line shows the file name (Figure 4).

Add my location (rounded to about 1 km): Tick this box to send your location with the photo. The browser may ask for permission first. When the photo already holds a location from the camera, the box reads **Location found in the photo** and stays ticked. Your tick carries over to the next photo until you tap **Reset**.

Save photo saves a copy of the original photo. On an iPhone, Save photo opens the share sheet; on a computer, Save photo downloads the photo.

Reset clears the photo so you can start over.

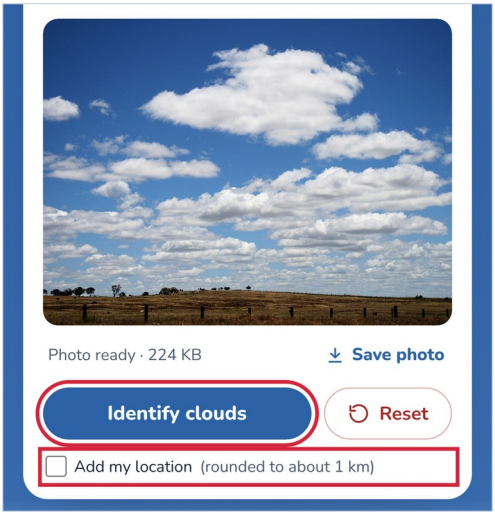


Figure 4. A photo ready to identify, with the Add my location option

For an example, the line under the photo reads **Example:** and the cloud name, and the Add my location option does not appear (Figure 5).

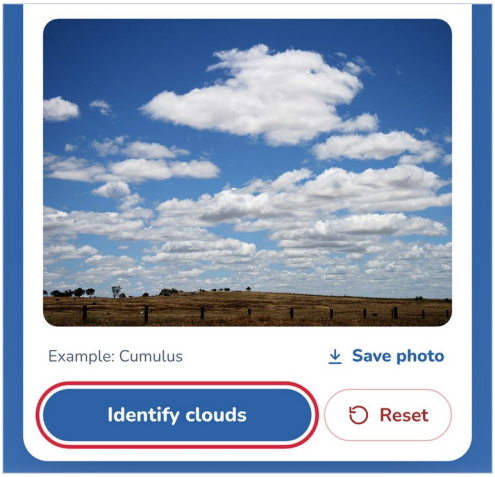


Figure 5. An example photo ready to identify

Step 4. Identify the clouds

Tap **Identify clouds**. The button changes to **Identifying…**, and the Results section reads **Reading the sky… usually 5 to 15 seconds** (Figure 6).

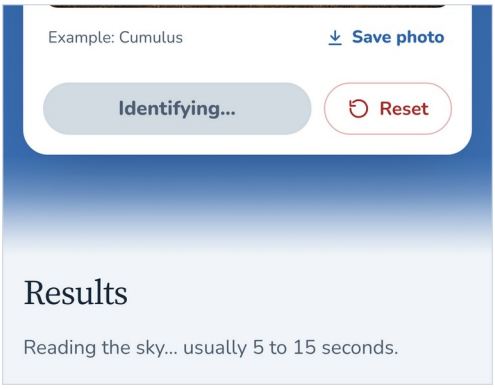


Figure 6. Identification in progress

Sometimes a Cloudflare check box appears that reads **Verify you are human**, and the Results section reads **Please tick the box above to continue** (Figure 7).

Tick the box, and the identification continues.

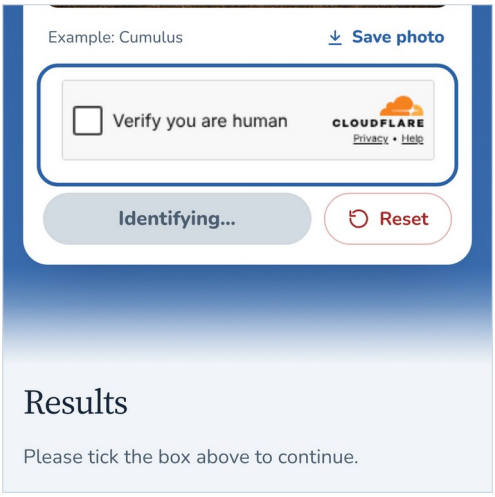


Figure 7. The Cloudflare check box

When the results appear, the button reads **Identified** and stays locked for that photo. To identify another photo, add a new photo or tap **Reset**.

Step 5. Read the results

Each cloud type found gets a result card (Figure 8). A result card shows:

- The level and altitude band, such as Low level · below 6,500 ft.
- The confidence, which is high, medium, or low, with three dots filled to match.
- The cloud name and what the cloud usually means.
- **Details for your brain**, which gives the Form and the Altitude.

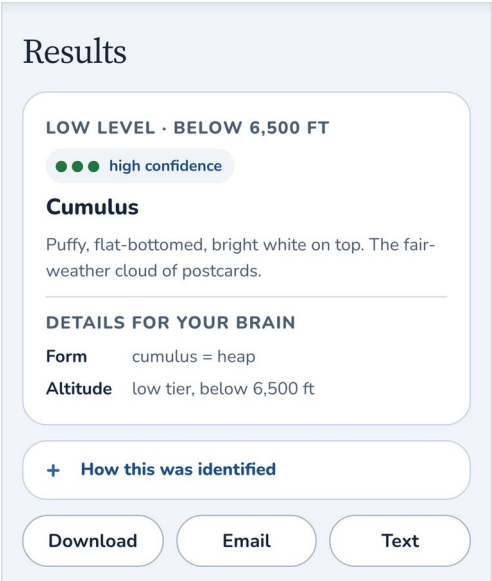


Figure 8. Result card for a cumulus photo

A photo with more than one cloud type gets one result card for each type, and each result card has its own confidence (Figure 9).

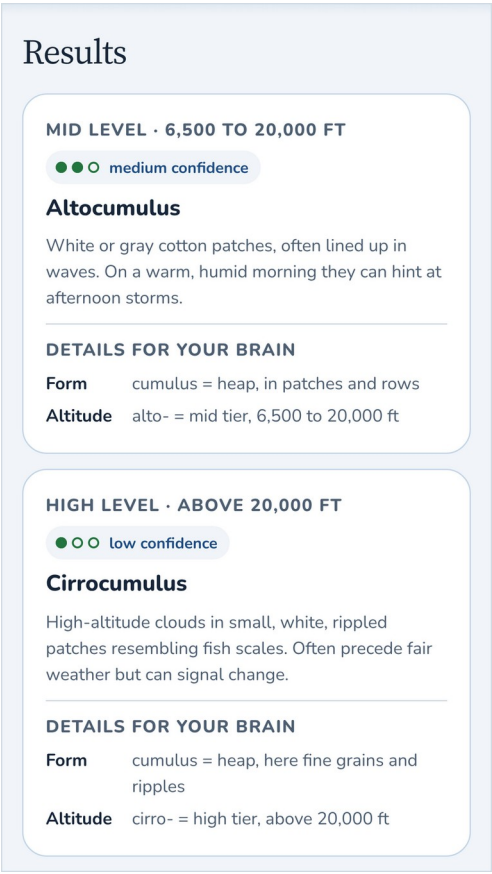


Figure 9. Two cloud types, with medium and low confidence

A fallstreak hole (hole punch cloud) gets its own **Special feature** card below the result cards, with its confidence and a short note about the feature (Figure 10).

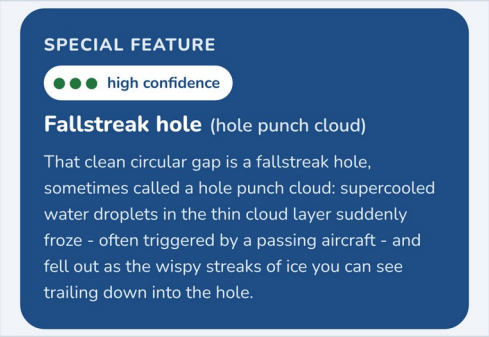


Figure 10. Special feature card for a fallstreak hole

When the photo shows no clouds, a **No clouds** card reads **Clear sky** (Figure 11).

A photo that does not show the sky gets a **Not a sky photo** card that reads **Try a wide shot of the sky**. A photo the site cannot use gets a **Photo not accepted** card that reads **Please choose a photo of the sky**. After either card, the Identify clouds button reads **Done**.

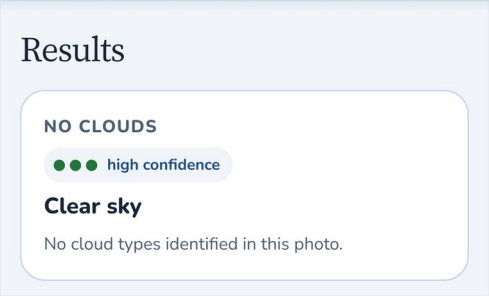


Figure 11. Clear sky result

Tap **How this was identified** to see the reasons behind the result, listed as **How high it looked**, **Texture**, **Light and shading**, and **Notes** (Figure 12).

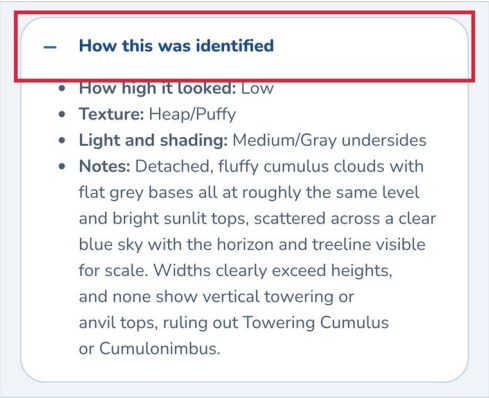


Figure 12. How this was identified, opened

Below the results, the **Clouds this tool can identify** list turns each cloud type found green, with a check mark (Figure 13).

Tap any cloud in the list to see a real photo of that cloud.

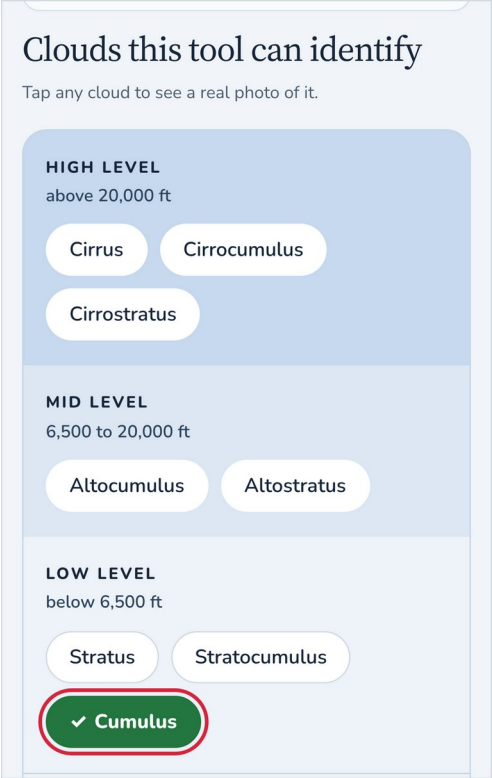


Figure 13. Cumulus marked green in the cloud list

The photo opens in a pop-up with the cloud's level, a short description, and the photo credit (Figure 14). Tap the **X** to close the pop-up.

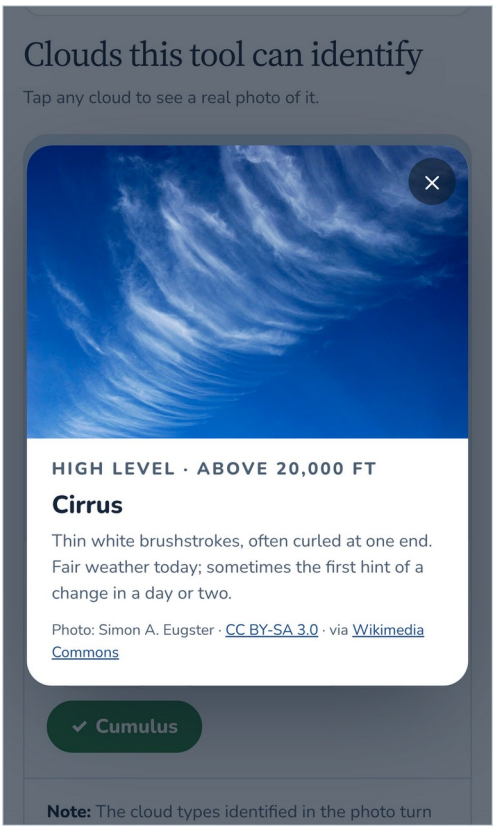


Figure 14. Photo pop-up for Cirrus

Step 6. Save, share, or report a result

Download opens the print dialog, where the result can be saved as a PDF (Portable Document Format) file.

Email and **Text** open the share sheet on a phone, with the photo attached. Where the browser cannot share a photo, Email and Text open a new email or text message with the results written in it.

Under the buttons, a photo of your own shows a line that starts **Thanks for contributing to science**, followed by the photo's eight-character archive number. An example shows a gray note instead (Figure 15).

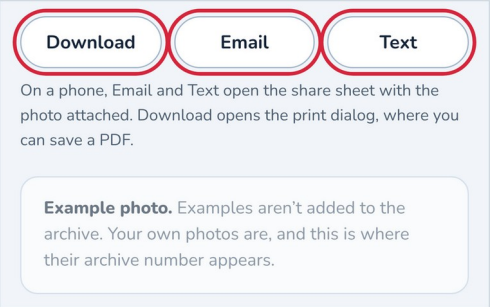


Figure 15. Download, Email, and Text, with the note for an example

Flag an issue: On a result for your own photo, each result card has a red flag button beside the confidence. On a computer, pointing to the flag shows the label **Flag issue**. Tap the flag to flag an issue with that result, and the site shows **Flagged**. Tap the filled flag again to take the flag back, and the site shows **Unflagged**. Example results have no flag button.

Start over: Tap **Reset**, or add a new photo.

Messages you might see

Message	What to do
Please tick the box above to continue.	Tick the Verify you are human check box above the Identify clouds button.
Not a sky photo: Try a wide shot of the sky	Take or choose a wider photo of the sky, then add it.
Photo not accepted: Please choose a photo of the sky	Choose a photo of the sky, then add it.
Couldn't read that one. Try another photo.	Try a different photo.
Sorry it took too long. Do me a favor. Try again.	Tap Identify clouds again.
Couldn't connect. Check your signal and try again.	Check the internet connection, then tap Identify clouds again.
Couldn't save. Try again. (beside a flag)	Tap the flag again.
Couldn't connect. Try again. (beside a flag)	Check the internet connection, then tap the flag again.

After an error message, the Identify clouds button unlocks, so the same photo can be tried again. After a Not a sky photo or Photo not accepted card, add a new photo.

The clouds DecodeClouds identifies

Group	Altitude	Clouds
High level	above 20,000 ft	Cirrus, Cirrocumulus, Cirrostratus
Mid level	6,500 to 20,000 ft	Altocumulus, Altostratus
Low level	below 6,500 ft	Stratus, Stratocumulus, Cumulus
Spans multiple levels	more than one level	Nimbostratus, Towering Cumulus, Cumulonimbus
Special feature	not applicable	Fallstreak hole (hole punch cloud)
No clouds	not applicable	Clear sky

DecodeClouds identifies 11 cloud types: the 10 standard World Meteorological Organization (WMO) cloud genera, plus towering cumulus. It also detects one rare feature, the fallstreak hole.

How each cloud type is recognized

DecodeClouds identifies clouds the way a trained observer does, following the World Meteorological Organization's International Cloud Atlas. It first records what it sees (how high the cloud looks, its texture, and how light passes through it) and only then names the type, testing the types in the order below, storm clouds first. For each type, the table gives what to look for, how the light behaves, and the sign that confirms it against its look-alikes. Learning these cues lets you read a result in the same language, explain why a cloud is what it is, and judge the answer for yourself.

Type	Look for	Light	Confirm
1. Cumulonimbus	A massive, towering storm cloud reaching from low level to very high altitude; the top flattens into a fibrous, cirrus-like anvil; a dark base, often with a rain shaft or a curtain of virga (rain that evaporates before reaching the ground).	Base dark to black and opaque; the sun usually blocked entirely.	Any anvil or fibrous, icy top means Cumulonimbus, never Cumulus. Brings lightning, heavy rain, gusts, sometimes hail.
2. Towering Cumulus	A very tall cumulus, clearly taller than it is wide, with crisp, bubbling cauliflower tops.	Brilliant white sunlit tops, distinctly darker flat bases.	Tops still hard-edged, not yet fibrous or flattened. The stage before a thunderstorm.
3. Cirrus	Thin, delicate white strands, plumes, hooks or bands high in the sky; mare's tails; detached threads.	Highly translucent; the blue sky stays bright behind it; no shadow on the ground.	No shading and no lumpiness: threads, not grains or heaps.
4. Cumulus	Detached, dense cotton-ball clouds with sharp outlines, rounded tops and flat bases all at one level; blue sky between them.	Bright white tops, flat grey bases.	Wider than tall. If it towers taller than it is wide, it is Towering Cumulus.
5. Cirrostratus	A thin, smooth whitish veil, fibrous or milky, often covering most of the sky.	The sun or moon shines clearly through it, often with a 22-degree halo; objects still cast shadows.	A halo, or a milky high veil with ground shadows intact.

Type	Look for	Light	Confirm
6. Altostratus	A continuous, featureless grey or blue-grey sheet.	The sun shows only as a dim glow, as through ground glass.	No halo and no sharp ground shadows: the two tests that separate it from Cirrostratus.
7. Nimbostratus	A dark, thick, continuous overcast; the day turned to gloom; often ragged shreds of cloud beneath the base; rain streaks, wet ground, a blurred dark horizon.	The sun completely hidden, its position not even guessable. Darkness separates it from Stratus and Altostratus.	A base that is diffuse and featureless. Rolls, lumps, ripples or waves in the base mean Stratocumulus, however dark the sky.
8. Cirrocumulus	A thin white patch or sheet of very small grains or ripples; a mackerel sky.	Pure white throughout, with no dark undersides.	Elements very small, about a finger's width at arm's length, judged only when the photo shows the horizon or ground for scale.
9. Altocumulus	Grey or white sheets and patches of rounded masses, rolls or parallel bands at mid level.	Distinct dark undersides and grey shadowing on the individual cloudlets.	Elements one to five degrees across (about three fingers at arm's length); the shading is the decisive test, size the cross-check.
10. Stratocumulus	Large, soft, clumpy rolls or patches, often with blue sky between; a thick layer can cover the whole sky and look as dark as a rain cloud.	Strong contrast: dark undersides and bright tops; in a full dark layer, a visible texture of lighter and darker patches.	Elements large (a fist at arm's length) when scale can be judged; a textured base separates it from Nimbostratus.
11. Stratus	A uniform, featureless grey layer like high fog; a flat, low ceiling; no rolls or lumps.	Heavily dims the sun, but if the sun shows, its outline stays clearly visible.	May bring mist or drizzle, not steady heavy rain; steady widespread rain means Nimbostratus.
Clear sky	The sky itself visible: blue by day or black with stars at night, the sun or moon sharp-edged if in frame, no grey or white layer anywhere.	A uniform grey, white or dull sky is always a cloud layer, never a clear sky.	A whitish veil with a halo is Cirrostratus, not clear sky.
Fallstreak hole	A sharply defined circular or oval gap in a thin layer of high or mid-level cloudlets, often with a tuft of ice or trailing fallstreaks in the middle; a long, narrow one is a canal cloud.	Not a cloud type: a feature within Cirrocumulus or Altocumulus. The hole forms where supercooled droplets suddenly froze, often where an aircraft flew through, and fell out as ice.	Reported only when clearly present, with its own confidence and a short note explaining it.

When two types look alike. Some pairs are often confused, so the identification applies a specific test to each pair. When the tool cannot settle a pair from the photo, it reports both types and lowers the confidence rather than guessing.

Pair	Test
Cirrocumulus or Altocumulus	Shading. Pure white cloudlets with no undershadows: Cirrocumulus. Distinct grey shading beneath: Altocumulus.
Cirrostratus or Altostratus	Halo and shadows. A halo, or a thin veil with ground shadows intact: Cirrostratus. A dim sun through a thick sheet, no halo, no shadows: Altostratus.
Altostratus or Nimbostratus	If the sun's position can still be made out: Altostratus. Fully blotted out, with steady rain: Nimbostratus.
Stratus or Altostratus	Stratus is low, with a sharply outlined sun through it; Altostratus is higher, with a blurred sun.
Cumulus, Towering Cumulus or Cumulonimbus	Vertical growth. An anvil or rain shaft: Cumulonimbus. Deep, hard-edged towers with no anvil: Towering Cumulus. Modest heaps wider than tall: Cumulus.
Cirrus or Cirrocumulus	Fibrous strands and hooks: Cirrus. Granular ripples: Cirrocumulus.
Altocumulus or Stratocumulus	Stratocumulus elements are markedly larger and darker, and the layer sits distinctly lower.
Stratocumulus or Nimbostratus	Structure, not darkness. Visible elements, rolls, ripples or waves and a distinct lower surface: Stratocumulus, even when dark and sunless. A diffuse, featureless base, often with rain streaks or ragged shreds: Nimbostratus.

How much to trust a result. The three confidence dots follow a fixed rule. High: the defining feature is plainly visible in the photo and no other type fits. Medium: the best fit, but one key check could not be made from this photo (for example the sun is out of frame, or the shot is too tightly cropped to judge the size of the cloudlets), or exactly one other type remains plausible. Low: two or more types fit about equally, or the photo is too cropped, dark or blurred to apply the tests. Open How this was identified under a result to see the observations the answer was built on.

Sizes are judged only when the photo includes the horizon, trees or buildings for scale; a zoomed patch of sky cannot show how big the cloudlets are, and the tool lowers its confidence instead of guessing.

Privacy and the archive

- Photos you identify, except examples, join the open DecodeClouds archive, a collection of sky photos for cloud and weather research. Archive photos are released under the Creative Commons Attribution 4.0 (CC BY 4.0) license.
- A location goes with a photo only when Add my location is ticked or the camera wrote a location into the photo. The location is rounded to about 1 km.
- To have a photo removed, email the photo's archive number to hello at decodeclouds.com. The **Contact** button at the top of the site opens an email to the same address.
- DecodeClouds is for educational purposes only. Cloud identification may be inaccurate or incomplete.

Photos shown in the screenshots: Cumulus by Toby Hudson and Cirrus by Simon A. Eugster, both under the Creative Commons Attribution-ShareAlike 3.0 (CC BY-SA 3.0) license, via Wikimedia Commons.